

Day 4 Component-Specific Troubleshooting

Practical

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseating Procedure

1. **Shut down the computer** completely and unplug the power cable.
2. **Open the computer case** and locate the RAM slots on the motherboard.
3. **Release the RAM module** by pressing the locking clips on both ends of the RAM slot.
4. Carefully **remove the RAM module** by holding it by the edges. Avoid touching the gold contacts.
5. **Inspect the RAM and slot** for dust, damage, or bent parts.
6. Align the notch on the RAM module with the key in the slot.
7. **Re-insert the RAM module** and press down firmly until the locking clips click into place.
8. Close the case, reconnect the power, and **start the computer**.
9. Check that the computer starts normally and that the RAM is detected.

Issues observed: No visible damage or dust was observed. The RAM module was successfully removed and re-inserted, and the computer started normally afterward.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

GPU Troubleshooting Checklist

1. **Identify the symptoms:** Check for no display, screen flickering, graphical glitches, crashes, or poor gaming performance.
2. **Check physical connections:** Make sure the GPU is properly seated in the PCIe slot and all required power cables are securely connected.
3. **Check the display:** Verify the monitor, display cable, and correct GPU display port are working properly.
4. **Check drivers:** Open Device Manager and update or reinstall the GPU driver if necessary.
5. **Check temperature:** Use monitoring software to check whether the GPU is overheating during use.
6. **Test the GPU:** Run a game or graphics benchmark and check for crashes, artifacts, or abnormal performance.
7. **Confirm the fix:** Restart the PC and test the system again to make sure the display and GPU performance are stable.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

I simulated a storage issue by shutting down the test computer and unplugging the SATA drive. After turning the computer on, the BIOS did not detect the storage drive, and the system displayed a **“No bootable device found”** error. The computer could not boot normally because the storage device was disconnected.

I then powered off the computer and securely reconnected the SATA data and power cables. After restarting, the BIOS detected the drive correctly, and Windows booted normally. I opened File Explorer and confirmed that the drive and its files were accessible, showing that the storage issue was resolved.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

I would check the **RAM first** because faulty or poorly connected memory can cause random restarts, crashes, and system instability. RAM is also easy to test and reseat compared with other components.

First, I would shut down the PC and reseat the RAM sticks properly. Then I would run **Windows Memory Diagnostic** to check for memory errors. If errors are found, I would test each RAM stick individually and try different RAM slots. Finally, I would restart the PC and run normal applications or games to confirm that the random restarts have stopped.